

Water Tunnel in a Box: Improvements to a Compact Flow Facility

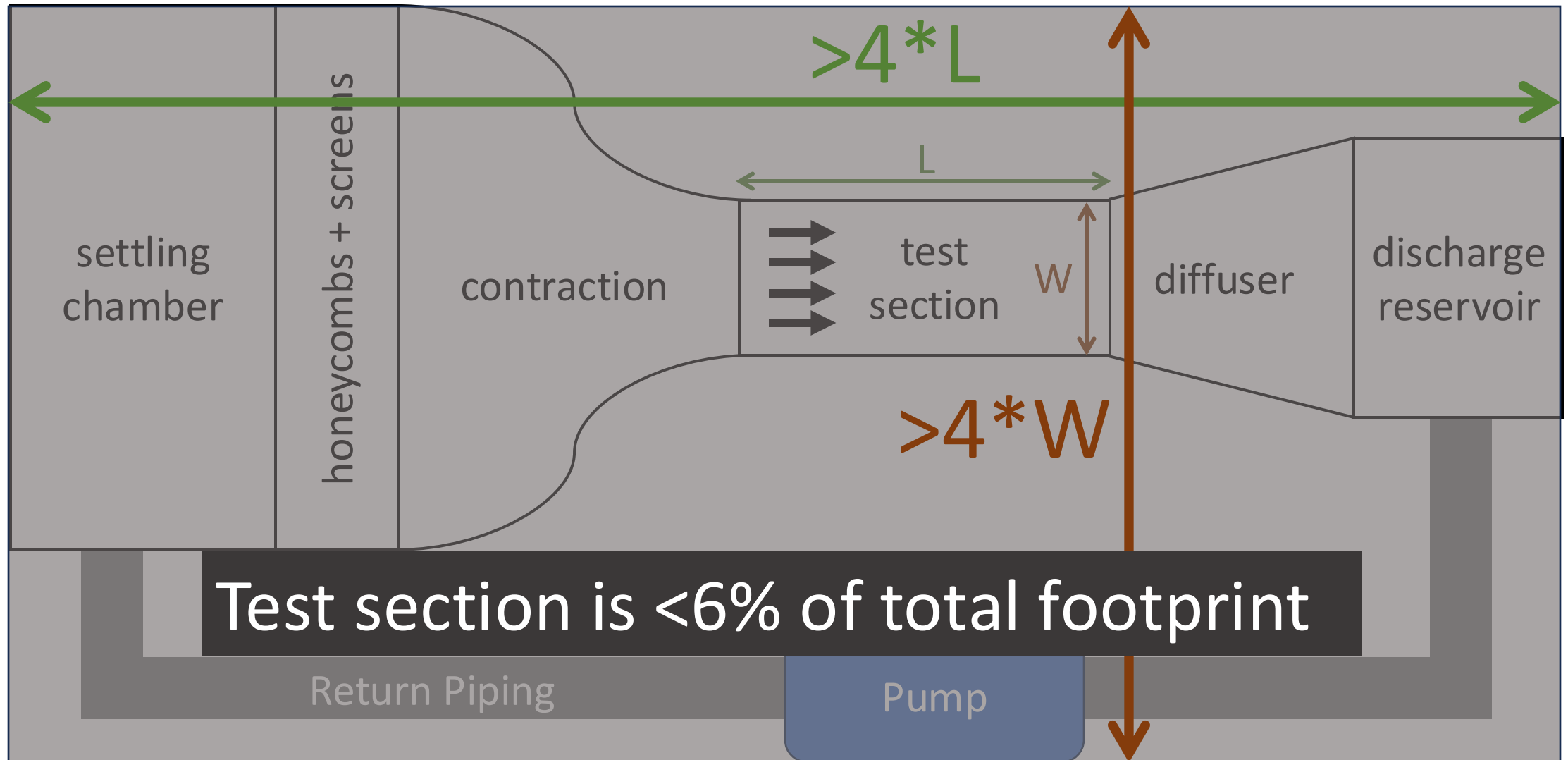
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SoCal Fluids XVII, UC Irvine

April 13, 2024

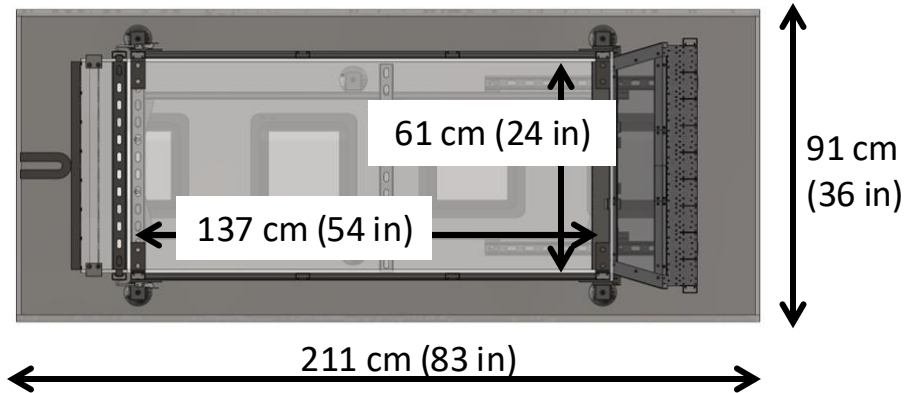
Supported by NSF GRFP under Grant No. DGE-1745301 and CAST.

Typical Water Tunnel Design

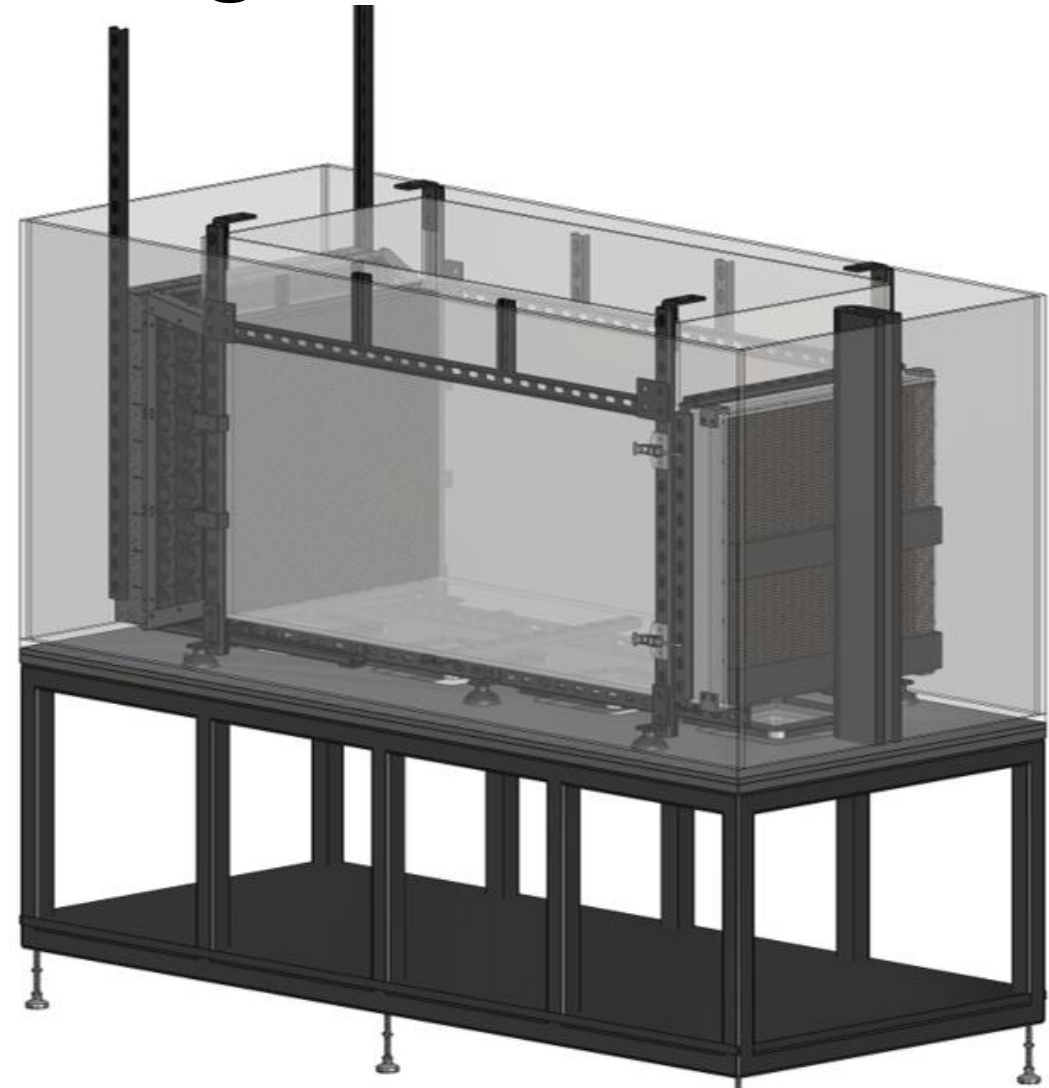
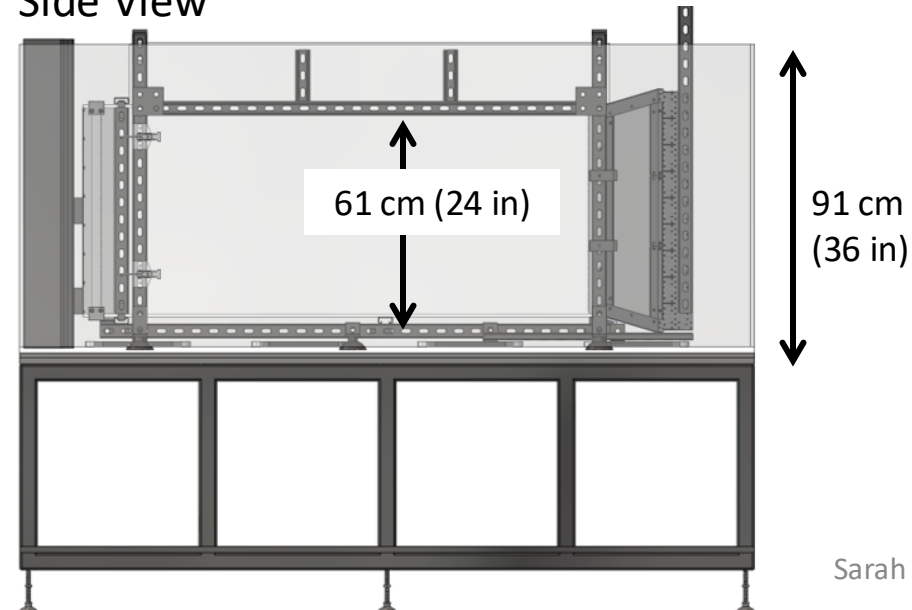


Water Tunnel in a Box Design

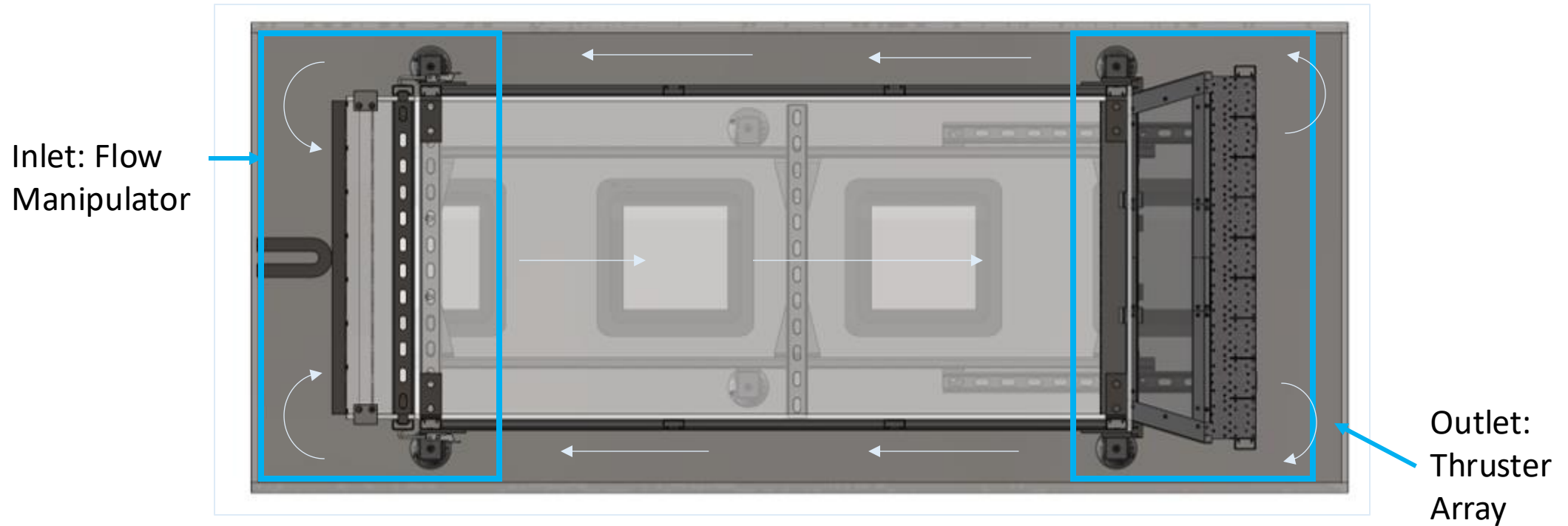
Top View



Side View

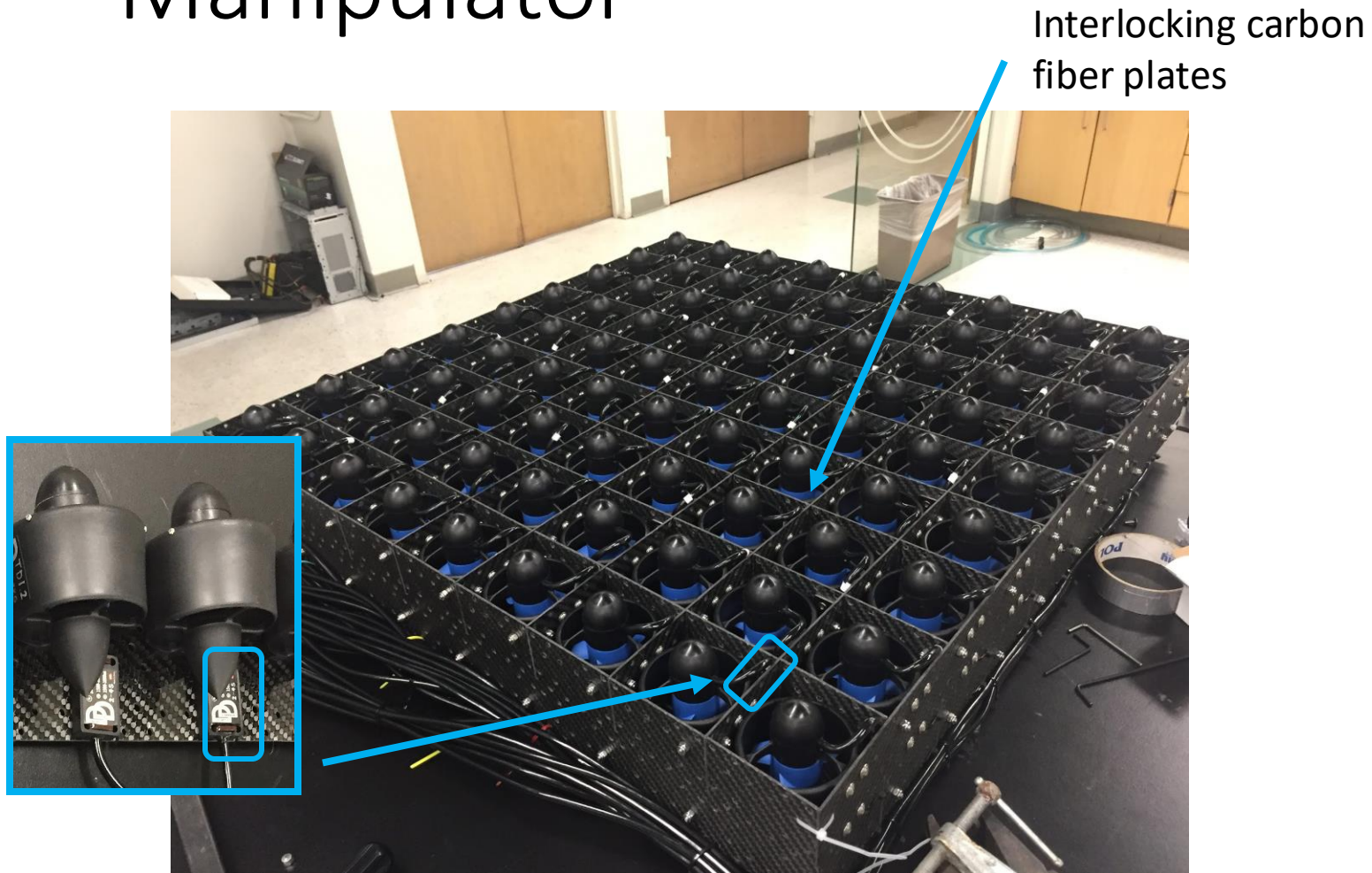


Water Tunnel in a Box Design



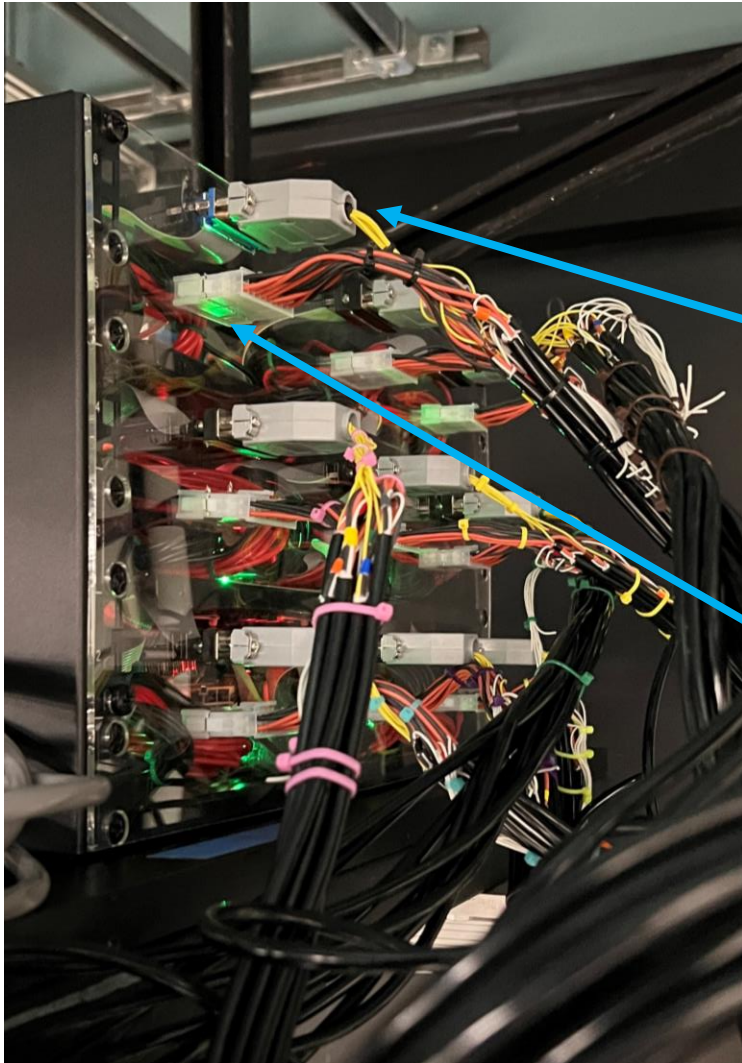


Thruster Array and Flow Manipulator



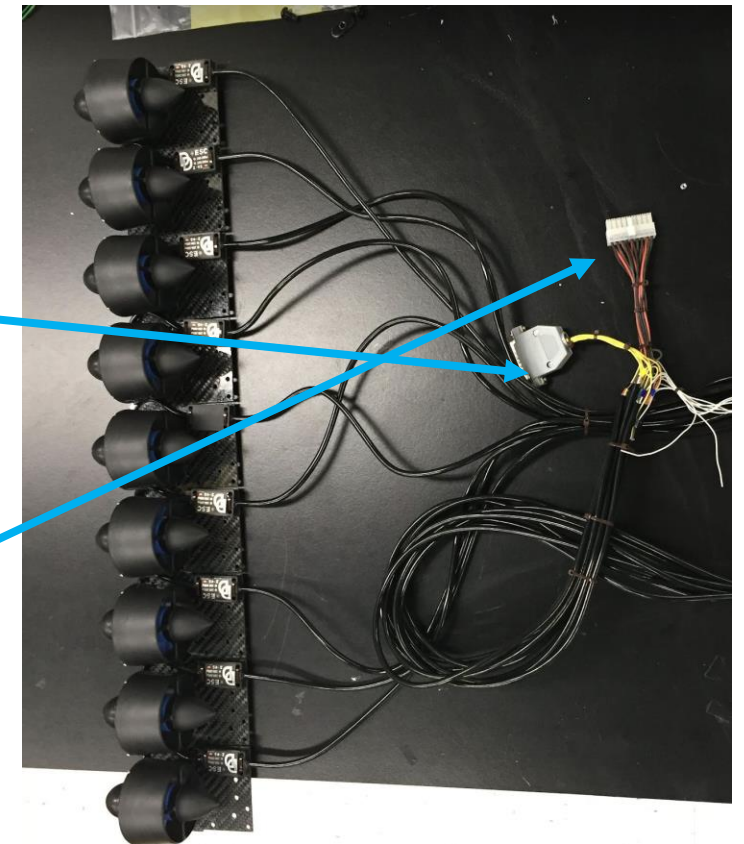
Flow manipulator

Wiring Configuration



Signal wire connection
(groups of 9 thrusters)

Power supply connectors
(9 thrusters to a single
power supply)

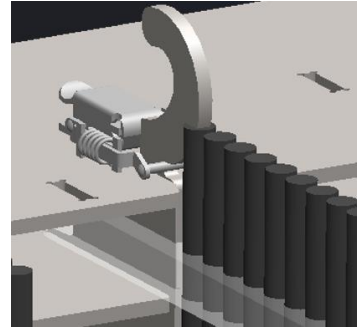
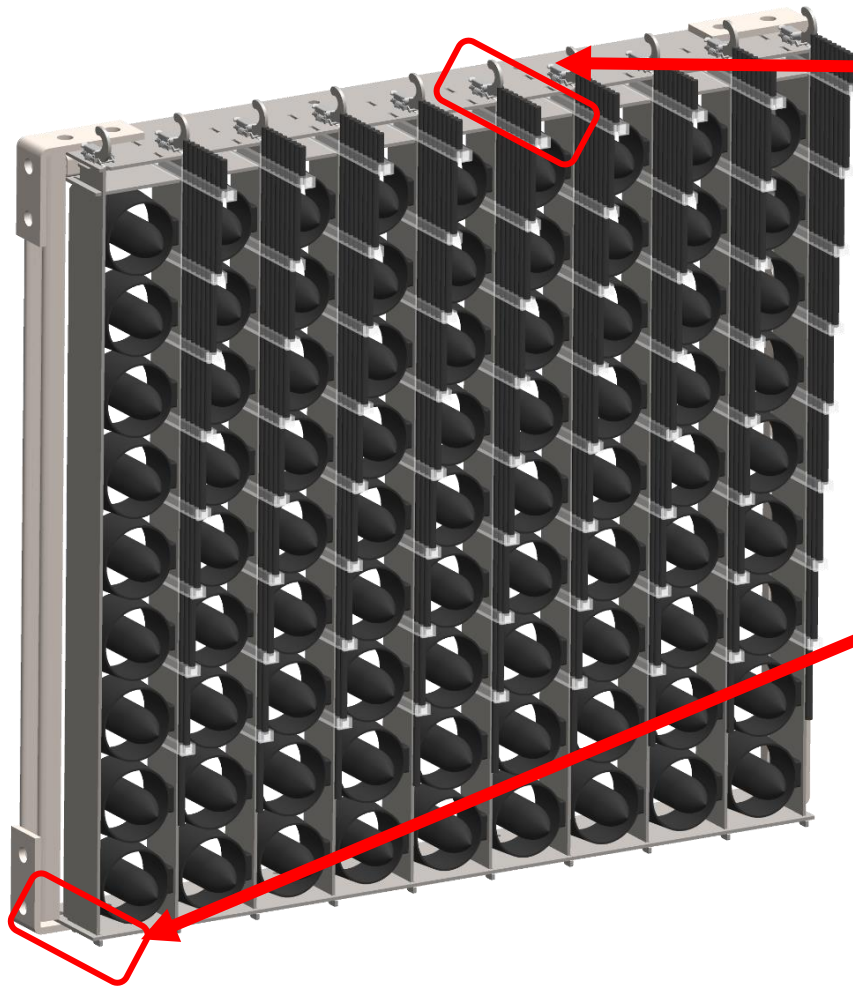


Design issues

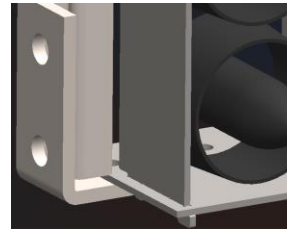
- Thruster lifespan of ~1-2 years requires replaceability
- Possible water damage to ESCs
- Labor intensive
- Hard to follow wire connections



Array Design Improvements

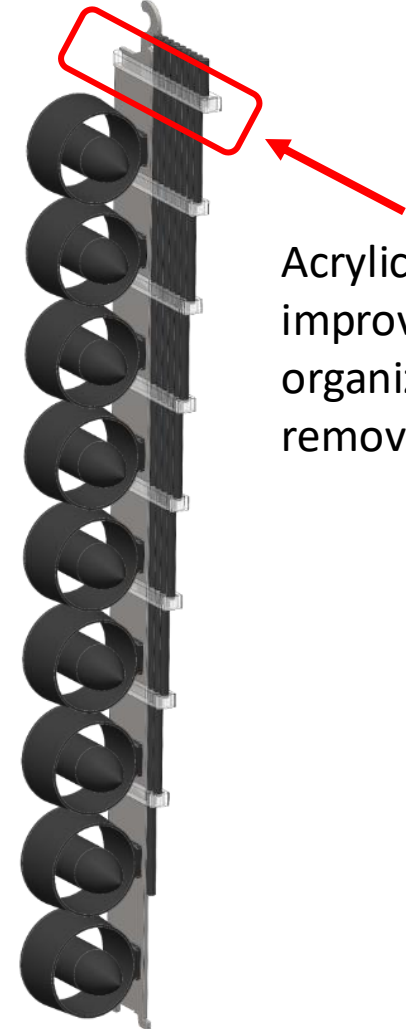


Hooks to secure in each column after sliding them in



The "foot" of each column slides into the bottom plate and locks in

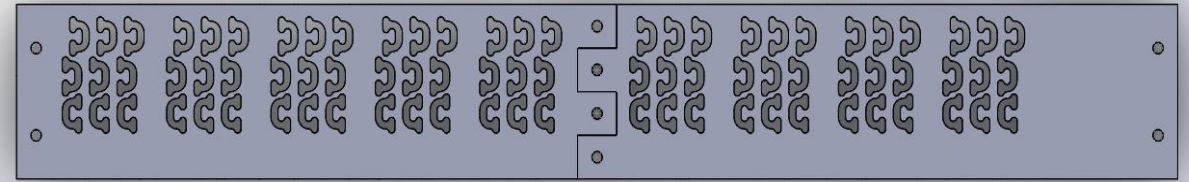
Modular Columns



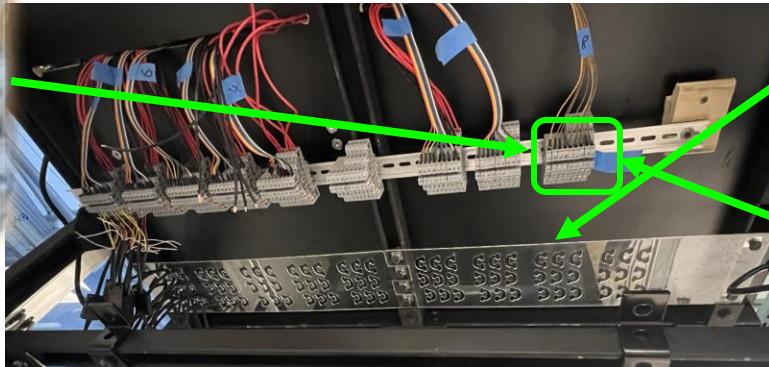
Acrylic clips for improved cable organization & removability

Wiring Improvements

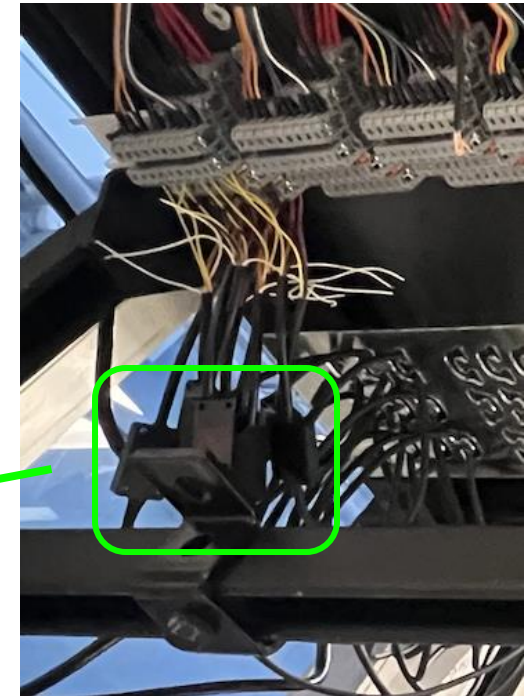
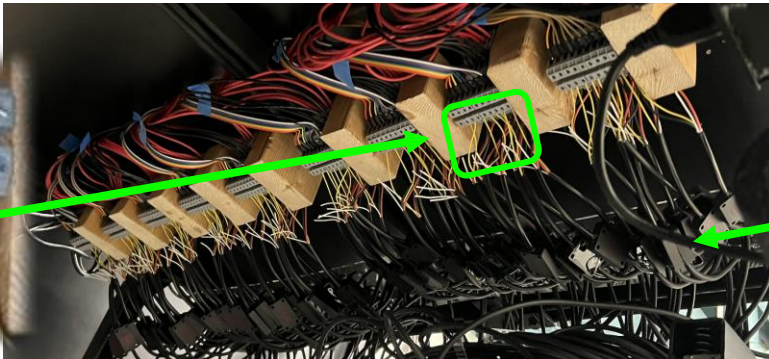
Plate to hold thruster wires



Wiring Changes

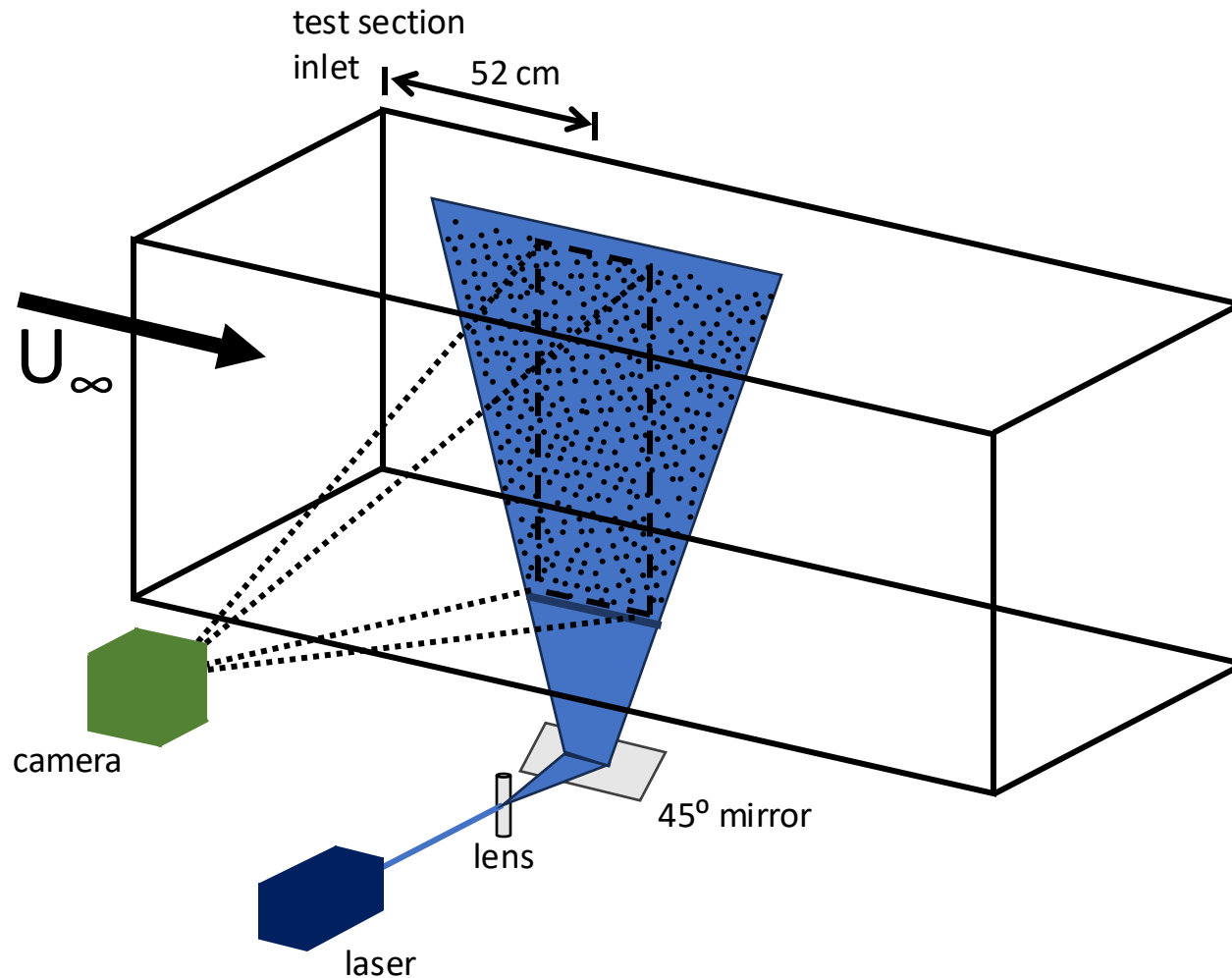


Screw terminal connections between signal, power, and ground wires



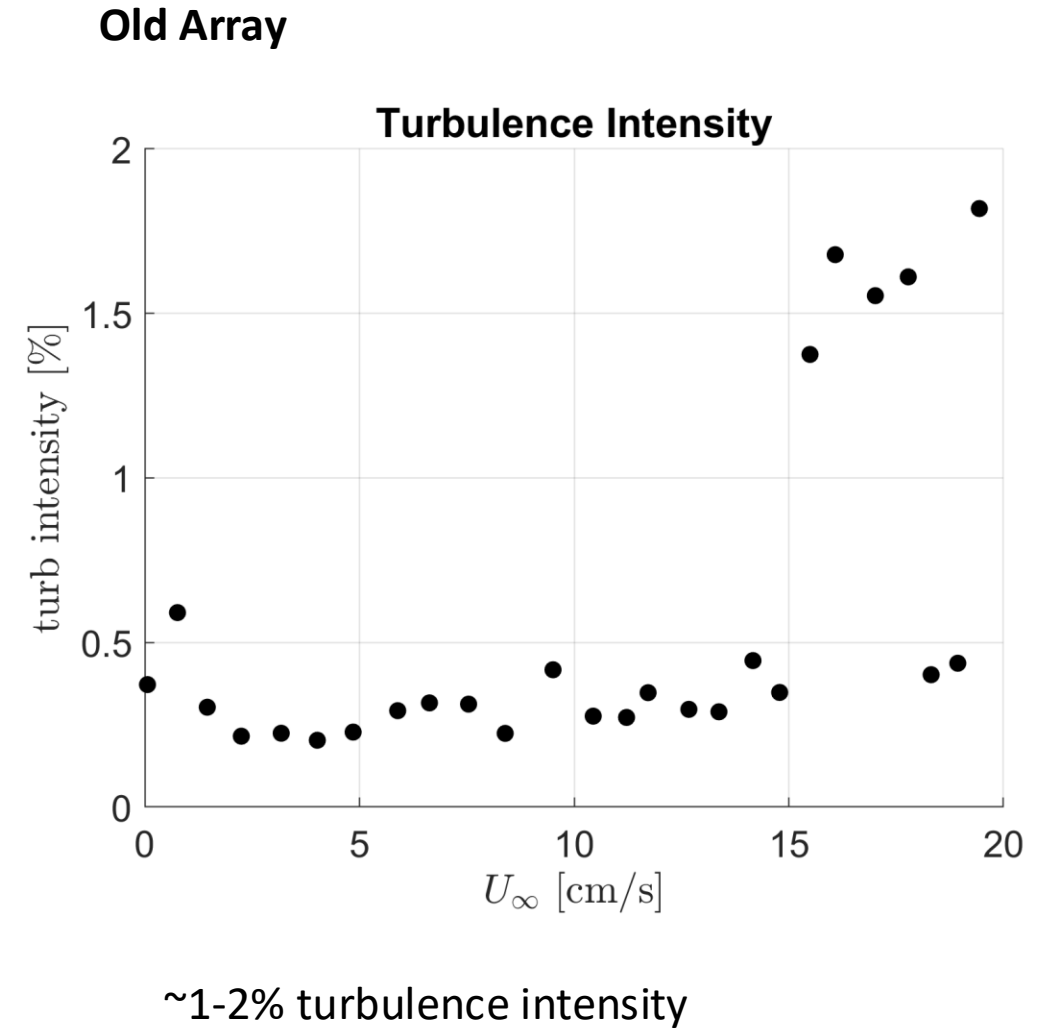
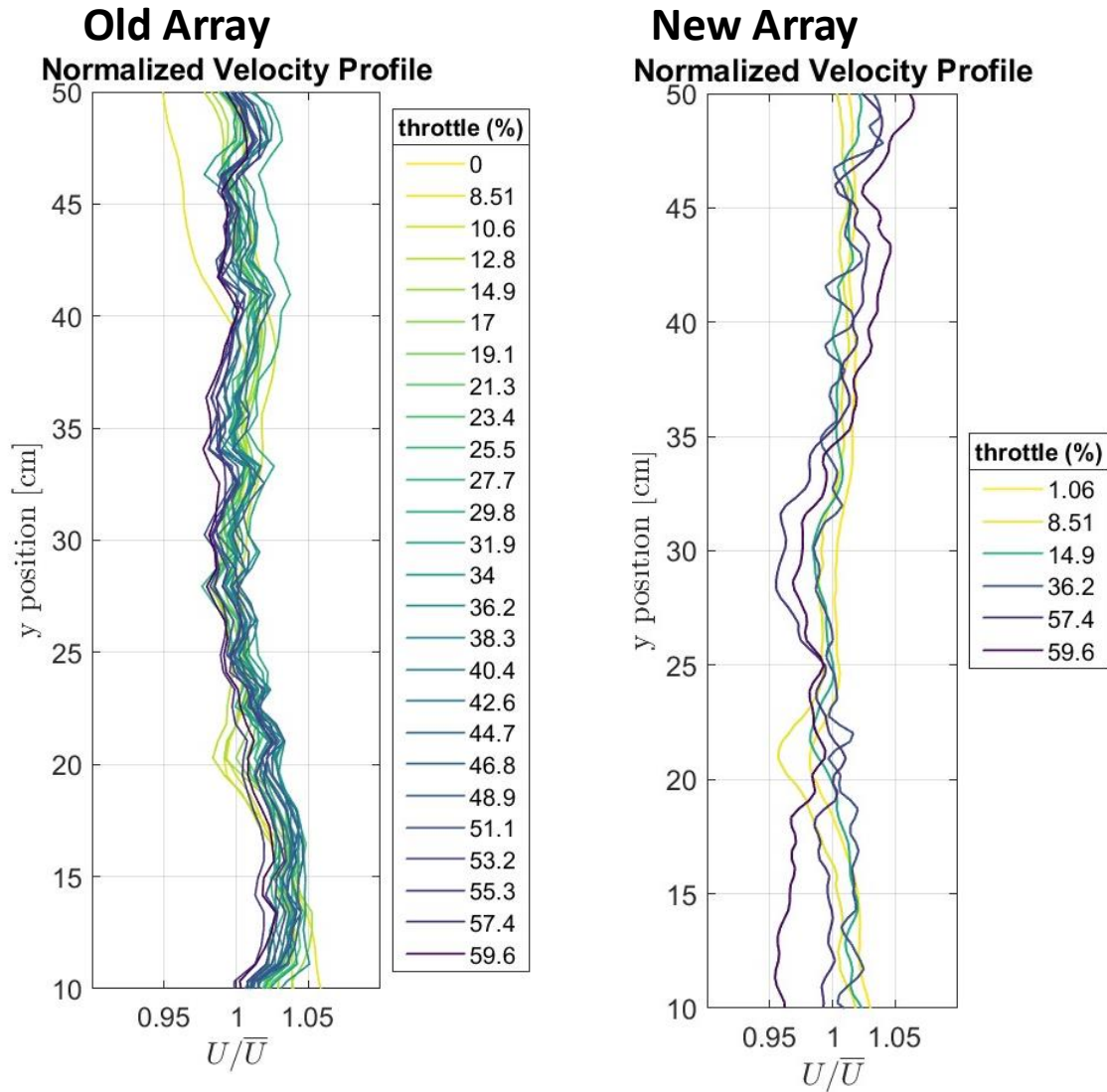
ESC's outside water near the power, preventing water damage

Tunnel Validation - PIV



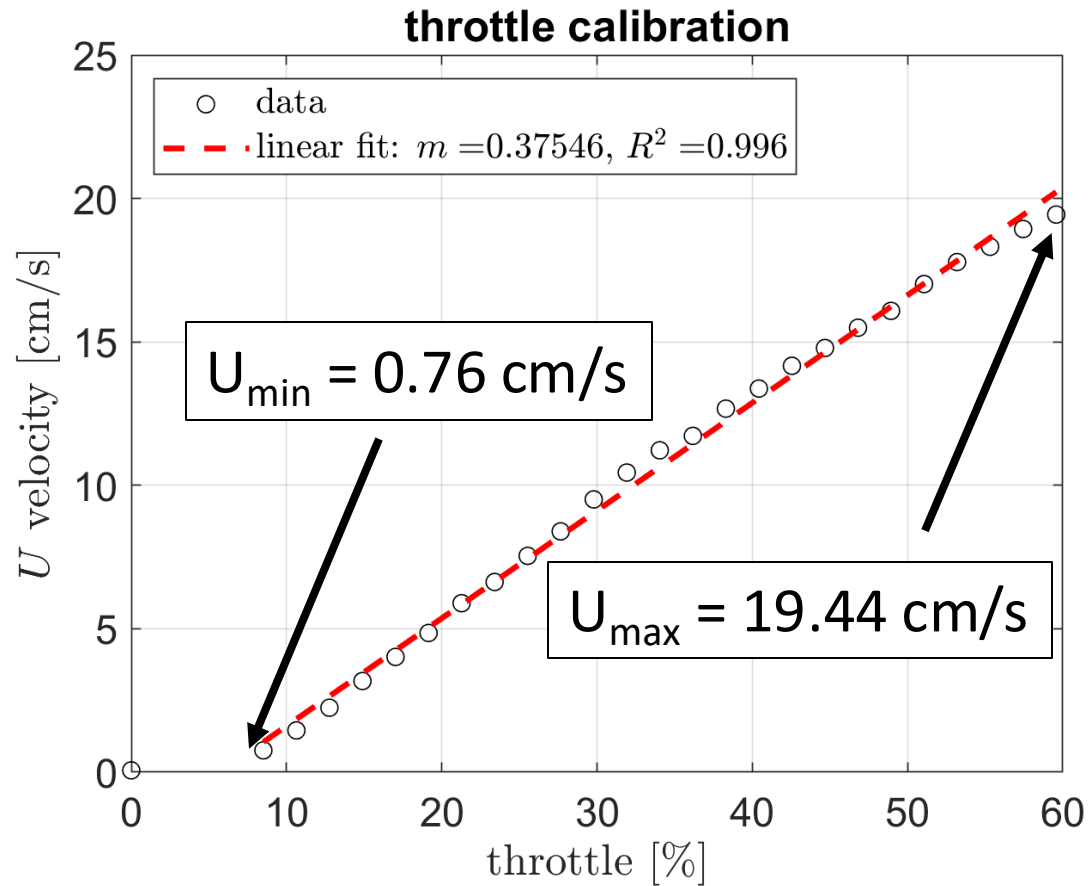
- Continuous laser
- Laser plane centered spanwise
- Time, streamwise averaged

Velocity profile data

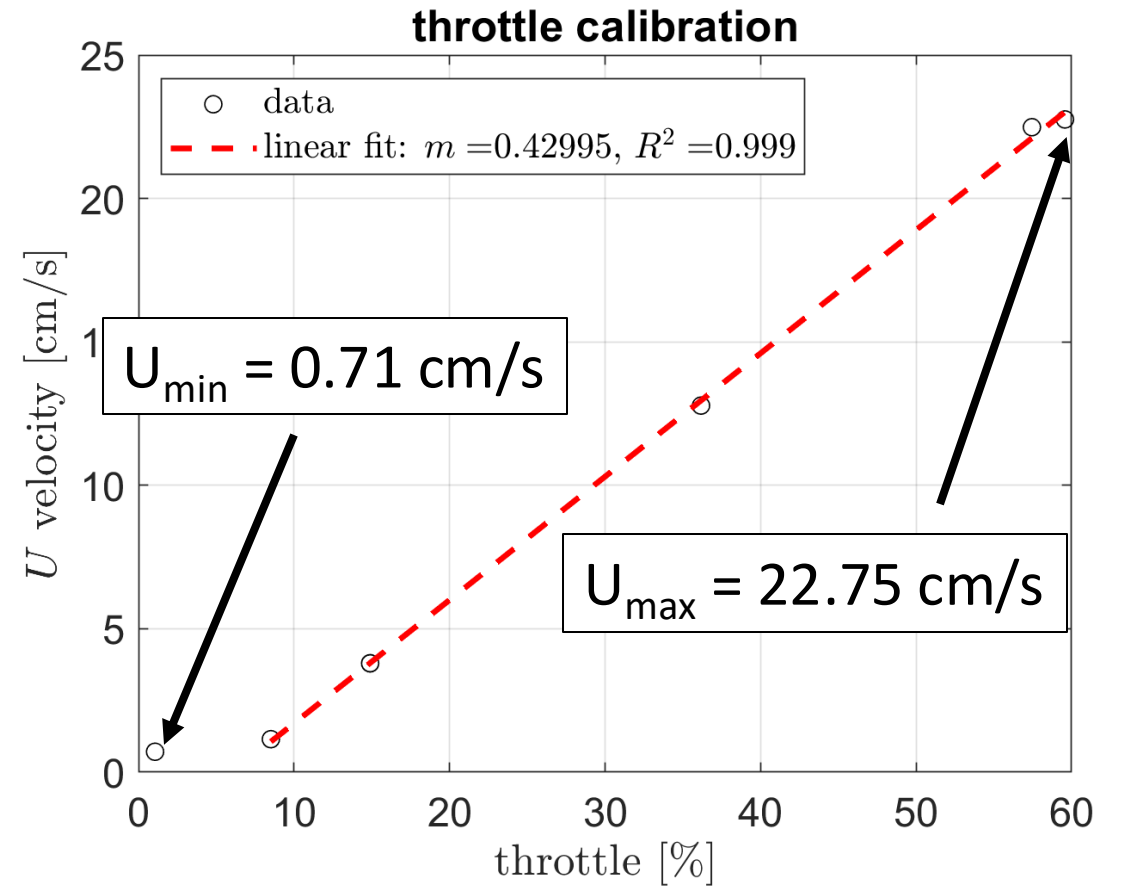


Velocity Profile Data

Old Array



New Array



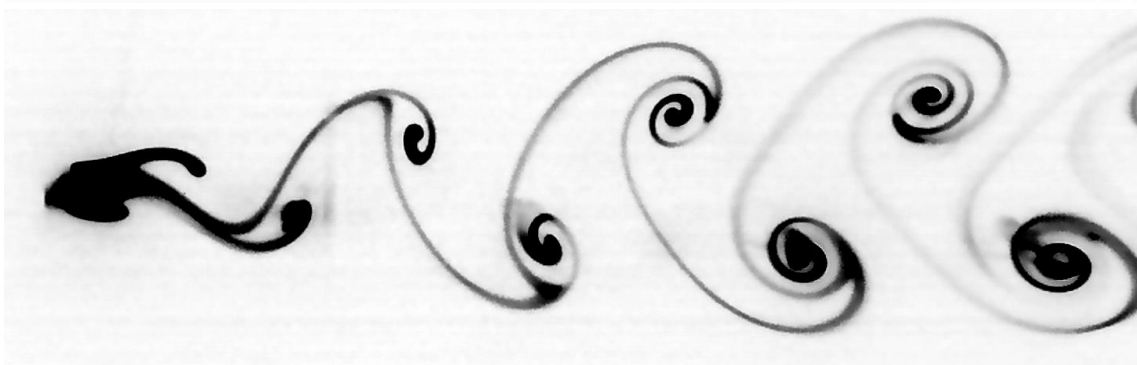
Re = 40



Re = 58



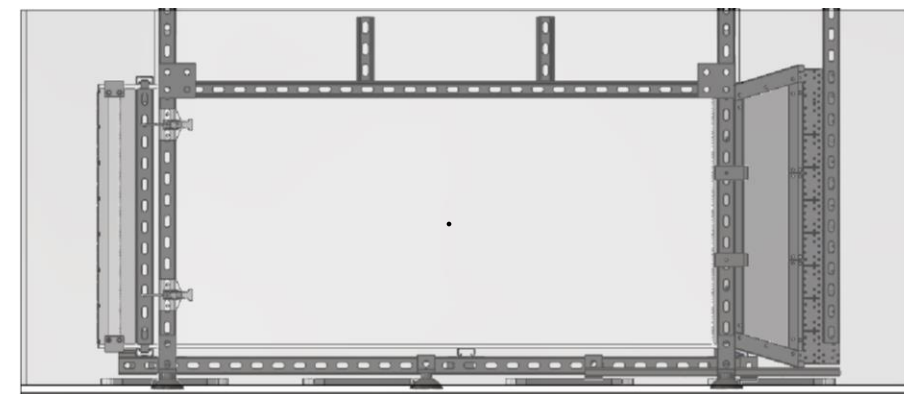
Re = 80



Re = 250



Re = 490

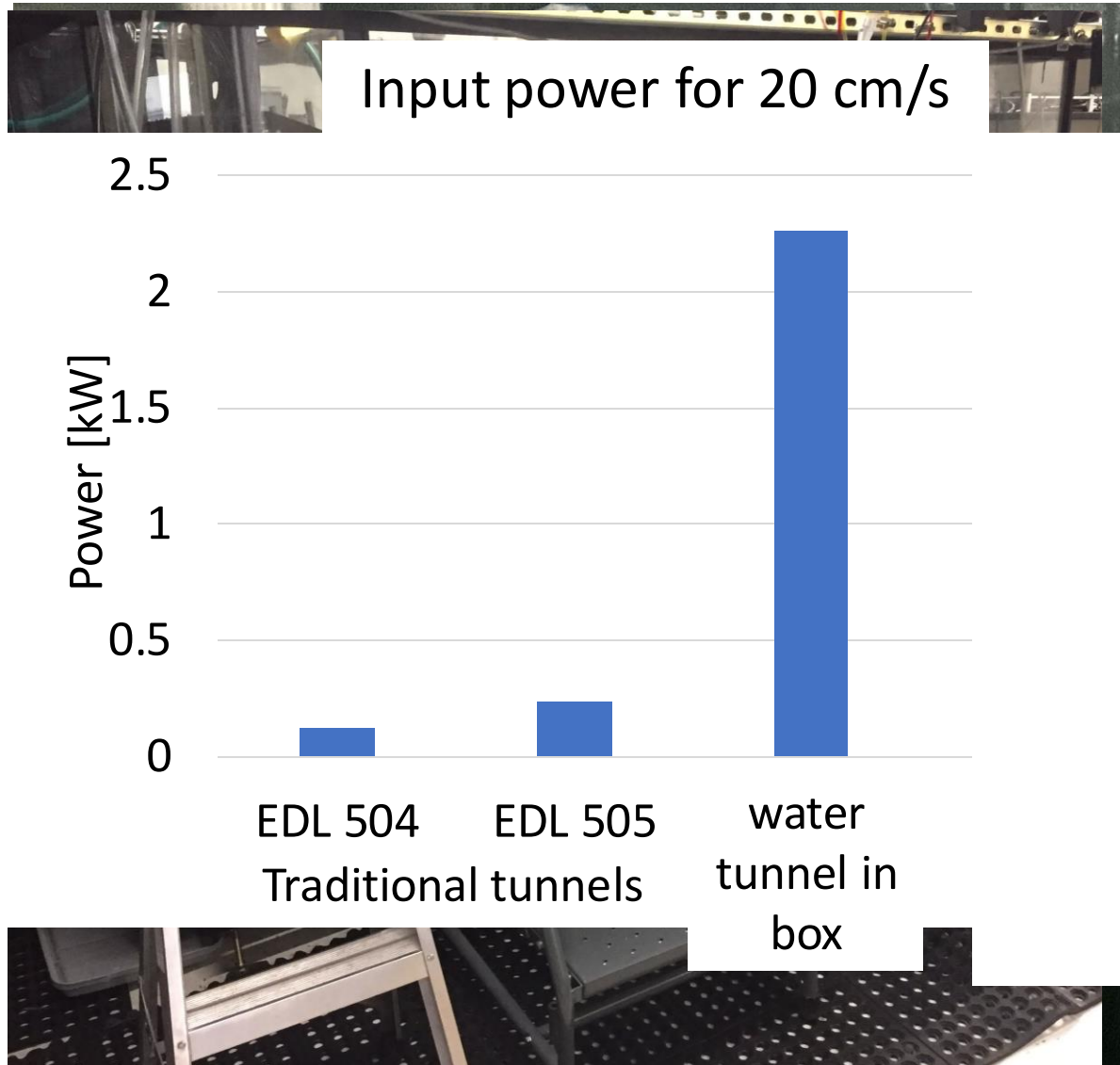


to scale

Cylinder diameter: 0.23" (5.9 mm)

Blockage ratio: 2.4×10^{-3}

Discussion



- Power tradeoff
 - insignificant at this scale
- Reduced total size:
 - faster acceleration
 - less slosh
- Thrusters offer flexibility:
 - Reverse operation:
 - Turbulence on demand
 - Shear flows
 - Spatially and temporally varying
- Simple geometry:
 - Inexpensive/rapid manufacturing
 - Access for cleaning/maintenance

Conclusion

- Test section volume 10x for same footprint
- Velocity remains within $\pm 5\%$ across test section
- Increased velocity range
- Future Work
 - PIV data in the crossflow plane
 - Turbulence measurements using miniLDV from MSE
 - Dye visualization

